CHEFS: Explanable DL and Edge-based Power Consumption Analysis Framework for Smart Homes

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***Abstract*—The growing use of smart technologies in homes has changed the way we manage power consumption. In this paper, *CHEFS* an innovative framework is proposed for smart home power consumption analysis, using Explainable AI (XAI) to improve clarity and accuracy with prediction and continuous decision-making by edge computing. Devices like the Internet of Things (IoT) and smart meters are used in smart homes to observe energy usage precisely. Such devices are also used for cutting energy costs and reducing environmental impacts. They often lack clarity in decision-making which makes it difficult, whereas the traditional machine learning models offer accurate predictions. To show this, we use XAI methods like SHAP and LIME to get a better understanding of the power usage patterns and forecasts. Edge computing reduces delays and bandwidth usage while also enhancing response time. This approach enhances clarity and dependability in Artificial Intelligence (AI) models providing reliable insights into household power consumption. A comprehensive evaluation helps create a more transparent and effective system for creating smarter energy solutions.**

***Index Terms*—Explainable Artificial Intelligence, Smart Home, Energy consumption, Edge Computing, Deep Learning, Machine Learning, SHAP, LIME**

I. Introduction

In conventional intelligent grid systems, consumers must pay a fixed power consumption charge for the units they use. Conventional smart grids did not offer any flexibility and security to the systems, which often led to power outrage, overconsumption or, sometimes, even blackouts. With the advancement in technology and the ongoing need for better services with minimized energy consumption, the concept of smart homes came into the picture. Smart homes are fully digitized and run by a centralized server system. Smart homes consist of smart meters that record and respond to real-time power needs and monitoring of the various systems. Smart homes allow homeowners to closely monitor their consumption patterns, leading to lower environmental impact and reduced bills. Smart meters have proved to be highly successful in the past few years as they offer people dynamic pricing, allowing consumers to pay for electricity based on the power units they have consumed at different rates. Integrating smart meters with homes also contributes to a better ecosystem. This has also reduced the load off consumers as smart homes work digitally which helps in automatic on and off load switch times of devices, enhances system security of homes using smart locks and motion sensors. Homeowners can manage their smart devices from anywhere using a smartphone or tablet. Smart home technology allows for customization based on individual preferences, enhancing personalized living experiences.

To monitor and ensure proper analysis, various Machine Learning (ML) and Deep Learning (DL) models have been applied by various authors. These models optimise energy consumption, fault detection, and predictive analysis. In [1], the authors provided a comparison between the previously used state-of-the-models and DL models such as Artificial Neural Network (ANN), Long short-term memory (LSTM) and Random Forests for electric load forecasting and load sharing. The authors also applied the models to demand response (DR) systems for analysis. A smart home energy management system (SHEMS) was implemented in [2], which implemented parallel processing, which is neurocomputing-based, and a forecasting mechanism for individual electrical devices. A multilayer perceptron model was implementation to check daily behavioural patterns which proved to be successful. A similar model implementation was carried out in [3], in which the authors used Smart plugs to measure Root mean square value of active, reactive power and other parameters. Hardwired using Arduino and Raspberry PI, the model showed accuracy of 97%.

In [4], the authors focused on the smart design model using the combination of DL models and Computer Aided Designs (CAD). Integration of these leads to improved identification accuracy, strong support for automatic control, and enhanced validity and security of models. In [5], a Feed-Forward Artificial Neural Network (FF-ANN) algorithm was proposed using a NoSQL database and a set of machine learning algorithms. The proposed methodology consisted of all the steps for Big Data analysis.

In correspondence to smart homes, IoT devices play a huge role in these. Internet of Things (IoT) devices are physical devices like home appliances and vehicles integrated with software technologies like sensors, motion detectors, etc. In smart homes, the various devices communicate with one over the network with minimal human intervention. This way, IoT devices help in collected real-time data for analysis and in optimizing energy usage.

While DL models provide accuracy and good predictions, they often fall into a ”black hole” after some time due to overfitting issues. This makes it challenging to arrive at conclusions , which is a drawback. For better transparency of the systems, we then employed Explainable AI (XAI) into the systems, which provides the ’why’ to consumers in better

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decision-making. XAI can explain why certain energy-saving recommendations are made or specific appliances consume more power. In [6], the authors reviewed Explainable AI and focused on Human-Computer interaction methodologies for better evaluation of energy consumption. To check the interpretability of Blackbox [7], the XAI techniques were employed using LIME and SHAP.

The models succeeded in giving consistent predictive current consumption patterns and energy usage. XAI was proposed in [8] to utilize electric load forecasting of a Norwegian residential building to minimise the energy consumption in Solar voltaic resources. Four different datasets were operated upon in [9] using XAI, Particle swarm optimization and XAI to check the utilization of energy resources. XAI made the most significant contributions in the smart grid domain. In [10], with the help of XAI, the model was able to extract essential insights and gather necessary specifications for developing a collaborative and human-centric system.

Existing smart home energy management solutions do not provide explainable information which makes it challenging for users to build trust and optimize their consumption patterns. To address this, we use XAI techniques to improve decision-making and transparency. Motivated by this, we present our *CHEFS* framework in our paper for Smart Homes.

A. Research Contributions

The significant contributions of this article are as follows.

- We proposed *CHEFS*, an intelligent framework that integrates XAI techniques to improve transparency and accessibility in predicting smart home power consumption usage. This is done by providing insights and emphasizing the factors responsible for every estimate generated by the regression analysis.
- *CHEFS* combines data collection and feature analysis from smart home devices and sensors, enabling real-time processing through edge computing.
- Further, the *CHEFS* is evaluated through various performance metrics such as training loss, RMSE, MSE, and MAE.

B. Paper Outline

The rest of the paper is organized as follows. The system model and problem formulation is elucidated in section II. The proposed *CHEFS* framework is shown in section III. Results are shown in section IV. The section V presents the conclusion and future work.

II. System Model and Problem Formulation

We propose a system using Explainable AI techniques to analyze and predict power usage in smart homes. The collection of system architecture of numerous components and variables is represented mathematically. Let H refer as the smart home environment consisting of several devices such as $\{d_1, d_2, \ldots, d_n\} \in D$. Each device is tracked and records its power consumption over time. Data points are recorded by the device $\{p_1, p_2, \ldots, p_k\} \in P$. Here each data point contains

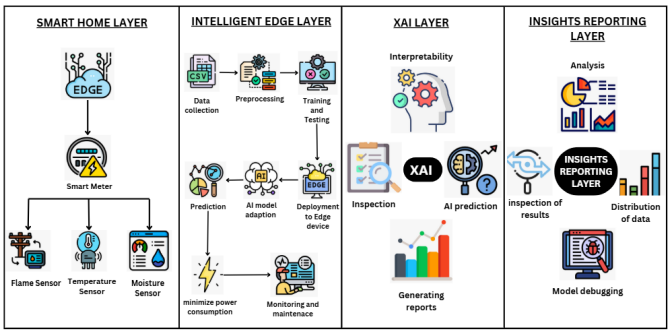

Fig. 1: *CHEFS*: The Proposed Framework

attributes such as energy used, time of usage, and operational status. Referring $P_i(t)$ as the energy consumed at time t by the device d_i, and let $P_{\text{total}}(t)$ denoting the sum of energy consumed by all devices at time t in the smart home. The aim of our system to is anticipate future power consumed and optimized energy performance. $\hat{P}_i(t)$ is represented as the predicted energy consumption is used in machine learning model M to predict energy usage of the device d_i. We use Explainable AI methods such as SHAP and LIME to give explanations for the predictions. This method is used to ensure transparency and understandability of our model.L denoting the loss function with A representing the accuracy rebukes incorrect predictions. The function F is created to minimise the loss and to increase the prediction accuracy.

$$F = \max_{M,H} (A - \lambda L) \tag{1}$$

where λ is a regularization parameter balancing accuracy and the penalty for incorrect predictions.

III. *CHEFS* - The Proposed Framework

The proposed *CHEFS* framework (as shown in Fig. 1) is organized into four subsections: the Smart Home layer, Intelligent Edge layer, XAI layer and Insights Reporting layer. Each layer is described as follows.

A. Smart Home Layer

The Smart Home Layer integrates a network of smart devices designed to deliver detailed insights into energy usage within a home. This layer comprises various devices, such as smart plugs, thermostats, lighting systems, and other appliances, all equipped with sensors to continuously monitor their operational status and energy consumption.

Smart devices gather real-time data on power usage, essential for analyzing energy dynamics. Let $P_i(t)$ denote the power consumption of device i at time t, expressed as:

$$P_i(t) = \beta_i \cdot U_i(t) \tag{2}$$

where:

- β_i is the device-specific power rating (in watts).

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- $U_i(t)$ represents the operational status or usage level of device i at time t (dimensionless or normalized).

The total power consumption of all devices in the smart home at time t is:

$$P_{\text{total}}(t) = \sum_{i=1}^{N} P_i(t) \tag{3}$$

where N is the total number of devices. For a time interval Δt, the total power consumption is calculated as:

$$P_{\text{interval}}(t, \Delta t) = \frac{1}{\Delta t} \sum_{k=0}^{M} \sum_{i=1}^{N} \beta_i \cdot P_i(t + k \cdot \Delta t_k) \tag{4}$$

where M is the number of discrete time steps, and $P_i(t + k \cdot \Delta t_k)$ represents the power consumption of device i at each step. This aggregated data, including timestamps and device identifiers, is stored in a CSV file, forming a structured repository for further analysis and model development.

B. Intelligent Edge Layer

The Intelligent Edge Layer integrates edge computing by using cutting edge technology advancements like Arduino or Raspberry Pi to process data collected by smart home devices. The edge device gathers data from various smart devices that are constantly gathering information about how much power users are using for research and safeguarding reasons. Since this approach reduces the amount of time required to collect and analyze data, it can lead to quicker decisions and responses, which is essential for reducing latencies. The data stored on the edge device is processed instantly. By maximizing usage of bandwidth through a reduction in the amount of data carried using cloud computing, and also reduces network traffic. Having better privacy and security inside the edge device is contributed by maintaining sensitive data on the network, also reducing the risk of data theft and controlling sensitive information. Moreover, the reliability is increased by the users while using the edge device as the essential function continue to process regardless in the case of unavailable internet access. The Raspberry Pi and Arduino are examples of edge devices that are used to design and execute such computations which enable AI-powered models to adjust to the particular needs and conditions of the smart home setting. This limited method with locally translated models reduces reliance on remote servers and speeds up smart home activities while yielding assessments which are remarkably precise and spatially sensitive.

1) Dataset Description and Preprocessing: The dataset used contains 17,520 rows and 14 columns. The dataset stores information on energy usage that a smart meter has gathered through different appliances in the home over time. Important features include dates and times including the power usage (in kW) of various smart home gadgets. The info-gathering provides comprehensive, time-stamped information about how each gadget affects the residence's total power usage. This information is an excellent tool for examining and improving trends in household electricity use. It is processed to prepare the data for examination and forecasting. Null and NaN values in the dataset were addressed by replacing them with the median values of their respective columns to ensure robustness against outliers and to mainatin integrity of real-valued features while integer columns were replaced using the mode for consistency. Let $\tilde{D}$ represent the dataset with m rows and n columns after preprocessing. Following the imputation of missing values, a group-by operation H was applied based on each unique timestamp t. This operation can be expressed as:

$$\mathcal{D}_t = H_t(\tilde{D}) \tag{5}$$

where $\mathcal{D}_t$ denotes the subset of $\tilde{D}$ corresponding to the timestamp t, and T represents the set of unique timestamps.

Subsequently, each subset $\mathcal{D}_t$ was divided into segments of size q according to the prediction intervals. This segmentation can be expressed as:

$$\mathcal{S}_{t,k} = \mathcal{D}_t\left[(k-1)q : kq\right] \tag{6}$$

where $\mathcal{S}_{t,k}$ denotes the k-th segment of size q from $\mathcal{D}_t$, and k indexes the segments within $\mathcal{D}_t$.

The resulting processed dataset $D \in \mathbb{R}^{m \times q \times n}$ was then formed by combining all segments across different timestamps. This can be represented as:

$$D = \bigcup_{t \in T} \bigcup_{k=1}^{K_t} \mathcal{S}_{t,k} \tag{7}$$

where T represents the set of unique timestamps, and K_t denotes the number of segments for each timestamp t. The Data preprocessing steps are shown in Algorithm 1.

Algorithm 1 Sequential Dataset Preprocessing

Input: Dataset $D \in \mathbb{R}^{m \times n}$
Output: Processed dataset $\mathcal{D} \in \mathbb{R}^{m \times q \times n}$
for $j = 1$ **to** n **do**
  if $\exists i$ such that $D_{i,j} = \varnothing$ **then**
    $D^{(j)} \leftarrow \text{fillNull}(D^{(j)}, \theta_j)$
    where $\theta_j = \begin{cases} \text{mode}(D^{(j)}), & \text{if } D^{(j)} \text{ is integer} \\ \text{median}(D^{(j)}), & \text{if } D^{(j)} \text{ is real} \end{cases}$
  end if
end for
Let $T = \{t_k \mid t_k \text{ is a unique timestamp in } D\}$
for $k = 1$ **to** $|T|$ **do**
  $D_{t_k} = \{D_{i,j} \mid \text{timestamp}(D_{i,j}) = t_k\}$
end for
Initialize $\mathcal{D} = \emptyset$
for D_{t_k} **for** $k = 1, \ldots, |T|$ **do**
  Partition D_{t_k} into segments of size q
  Append each segment to $\mathcal{D}$
end for
return $\mathcal{D}$

2) Working of ANN: Artificial Neural Networks (ANNs) is an architechture with several layers of connected neurons that simultaneously process input data and predict accurate outputs. The accuracy of these outputs depends on weights, due to which the model incorporates a bias value and then uses a non-linear activation function to produce its output. Thus, the values of weights and bias can be modified during the training, which is a major advantage and allows more accurate results

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and enables the ANN to identify complex patterns and links in data. The output o_m^n of a neuron m in layer n of an ANN is computed as follows:

$$o_m^n = \phi\left(\sum_{r=1}^{Q_{n-1}} \omega_{mr}^n o_r^{n-1} + \beta_m^n\right) \tag{8}$$

where o_m^n is the activation of neuron m in layer n, ϕ represents the activation function, ω_{mr}^n is the weight connecting neuron r in layer $n-1$ to neuron m in layer n, o_r^{n-1} is the activation of neuron r in the preceding layer, β_m^n denotes the bias term for neuron m in layer n, and Q_{n-1} is the number of neurons in the preceding layer. Equation (8) shows the process of ANN transformation of the data from the previous layer by neurons present in each layer, using learned weights and biases to produce activations that represent higher lever notions of the input data. The factors of ANN have been refined using techniques such as backpropagation and other streamlined algorithms to increase predictive clarity.

C. XAI Layer

Here, Explainable AI methods like SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) are essential for clarifying how sophisticated machine learning models make decisions. Originating in cooperative game theory, SHAP values provide a constant measure of an attribute's importance by assessing each characteristic's contribution to a given prediction. These technique provides both global and local comprehension by equally spreading the total anticipated value among qualities. The predictions are affected by the difference in the value of the features. The reason behind this is shown by SHAP whereas the LIME gives priority for approximating the complicated model around a particular prediction by local comprehension with a straightforward approach. This is achieved by firstly modifying the input data which examines how the estimate changes and then fits a simple model to the altered points. Of the resultant LIME explanatory graph, the characteristics that are favorable are shown on the right in the orange, whereas those that contribute adversely are shown on the left in the blue. This graph depicts the most important characteristics of the given prediction. These techniques enhance the accessibility and transparency of algorithm activities when combined. This facilitates a deeper understanding of the model's outputs.

D. Insights Reporting Layer

This layer seeks to transform the comprehensive energy use statistics that the smart meter gathers through multiple smart home gadgets into useful information. The building serves as the primary evaluation point across all of the smart meter's power use data. The descriptions produced in this section, which compile the information gathered through the XAI models, provide clear and useful assessments of the household's power usage. These reports give users the information they need to arrive at informed choices about optimizing energy efficiency. They include descriptions and images that help users understand the ways that various devices affect overall utilization. Rapid and immediate reporting is made available by Edge Computing, ensuring that customers receive accurate and up-to-date data that is tailored to their particular household power patterns.

IV. Result Analysis

This section shows *CHEFS* framework's performance analysis using various statistical metrics parameters such as training loss, RMSE, MSE and MAE.

A. Experimental Setup and Tools

This experimental setup of the paper uses high-end processors with a server having an x86-64 architecture, two T4 GPUs, 12 GB of RAM, and 108 GB of hard disk space. For faster calculations of data and encapsulation of numerical functions we used Numpy and PyTorch ML framework. We also used various libraries like GitPython-3.1.42, datasets-2.18.0,dill-0.3.8, docker-py creds-0.4.0, gitdb-4.0.11, multiprocess-0.70.16.

B. Performance Analysis of the CHEFS Framework

1) Training Loss: The model's learning process over 1000 epochs is shown in Fig. 2a. Initially, within the first 200 epochs there is a rapid decline in loss from approximately 0.025 to below 0.005. This shows effective learning and parameter optimization. The loss further continues to reduce as training moves forward with slight fluctuations around the 400th epoch. This indicates instability. Regardless of these oscillations, the loss steadily converges to a stable with the overall trend remaining downward with a stable value of about 0.003. This shows the model continues to fine-tune its parameters and reach a more ideal state with improved generality.

2) RMSE: The graph of RMSE in Fig. 2b starts at around 0.16 and quickly decreases during the first 200 epochs, indicating significant improvements in the accuracy of the model. Between the 200th and 400th epochs, smaller fluctuations are observed, that gradually decrease and stabilize beyond 400 epochs around 0.04.This consistent reduction in RMSE demonstrates the model's effective learning and ability to generalize better over time.

3) MSE: The graph in Fig. 2c represents the MSE values over 1000 epochs. Starting with a sharp drop in the first 200 epochs around 0.025. This indicates rapid learning and parameter optimization. Due to the model refining its predictions, the MSE reduces slowly from 200 to 600 epochs. After 600 epochs, the MSE stabilized with minor fluctuations of around 0.003. This suggests that the model has converted largely and making small progress with further training. Altogether, MSE reflects the model's accuracy over time.

4) MAE: The graph in Fig. 2d shows the MAE over 1000 epochs. At first, within the first 200 epochs, the MAE experiences a steep reduction from around 0.055 to 0.035. This indicates significant model learning at the early stages of training. It continues to reduce at a slower and more irregular rate fluctuating around 0.025 and 0.030. These oscillations show periods of minor overfitting throughout the process of training. Although the entire trend continues to stay stable,

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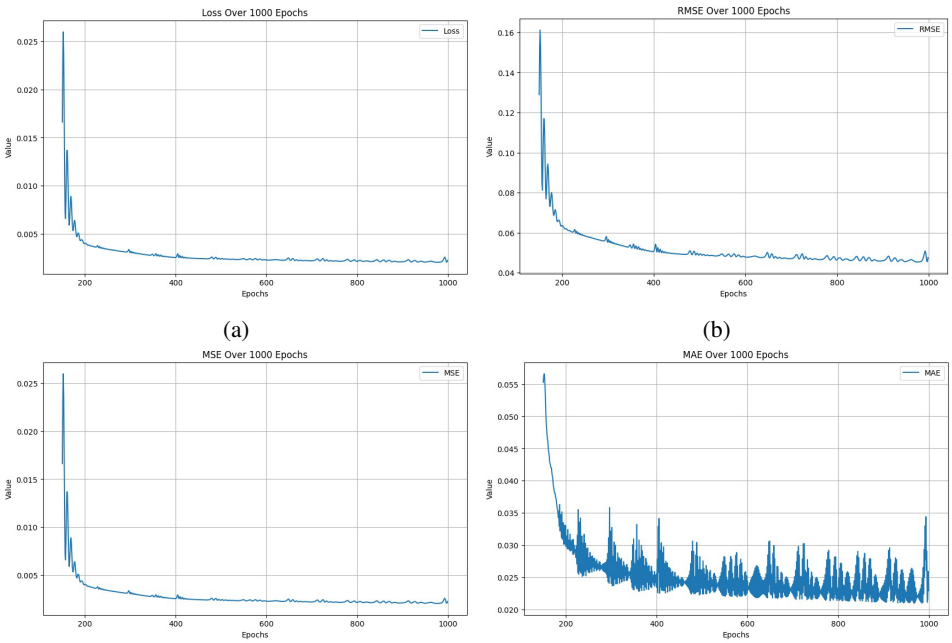

(a) (b) (c) (d)

Fig. 2: Performance analysis of ANN model (a) training loss, (b) RMSE, (c) MSE, and (d) MAE.

the model sustains consistent performance regardless of the fluctuations.

C. SHAP Interpretation

Fig. 3 shows the three SHAP plots that depict the contributions of individual features in the model's predictions for different instances. In Fig. 3a, the baseline prediction $E[f(X)] = 0.838$ is adjusted to a final output of $f(x) = 0.84$, with Feature 6 making the most positive contribution of 0.21, while Feature 1 has the largest negative impact of -0.08. In fact, other features like Feature 9 and Feature 2, which had a negative influence, are still contributing less; hence, the final prediction is slightly above the baseline due to the strong positive effect of Feature 6. Fig. 3b shows a final prediction of $f(x) = 0.825$ from the same baseline. Here, Feature 11 contributes the most positively with 0.37, but Feature 1 has the most significant negative impact of -0.17. The final prediction is slightly below the baseline, primarily due to the stronger negative effects of several features, even though Feature 11 provides a notable positive influence. In Fig. 3c, the final output is $f(x) = 0.807$, with Feature 11 again showing the strongest positive impact of 0.4, while Feature 1 contributes the most negatively with -0.17. The final prediction remains below the baseline due to the cumulative negative effects from features like Feature 9 and Feature 6, despite the considerable positive influence of Feature 11.

D. LIME Interpretation

Fig. 4 provides insights into how individual features influence the model's predictions across different instances. In Fig. 4a, Feature 1, with a high value (greater than 0.31), is the most significant factor that contributes positively to the prediction, while Features 9 and 6 also positively impact the outcome. Conversely, Feature 11 (with values between 0.04 and 0.11) and Feature 5 have a negative impact on the prediction. This implies that higher values in Features 1, 9 and 6 contributes positively to the prediction score, while specific ranges in Features 11 and 5 reduce it, highlighting the key factors influencing the model's decision. In Fig. 4b, Feature 7 (with values greater than 0.03) is the most decisive one in moving the prediction towards the positive class, whereas Feature 11 (with values less than 0.04) strongly pulls the prediction towards the negative class. Features 8 and 3 are the positive class contributors up to a certain threshold and beyond.In contrast, Features 6 and 9 are the ones that negatively influence the prediction within specific value ranges. Finally, Fig. 4c demonstrates that Feature 1 (with values greater than 0.31) is the main reason for positive predictions, which is significant with a value of 0.44. Feature 11 (with values above 0.35) and Feature 6 (with values greater than 0.13) are also positively contributing with 0.35 and 0.26 respectively. Feature 7 (with values above 0.03) and Features 10 and 3 (with values above 0.04 and 0.01, respectively) contribute more moderately to positive predictions. On the negative side, Feature 9 (with values between 0.05 and 0.06) and Feature 4 (with values between 0.05 and 0.14) contributed to the predictions with values 0.09 and 0.04 respectively.

V. Conclusion

In conclusion, *CHEFS* framework presents a transformative approach to energy management by integrating XAI and Edge Computing for power consumption analysis in smart homes. This research highlights the critical role that XAI plays in

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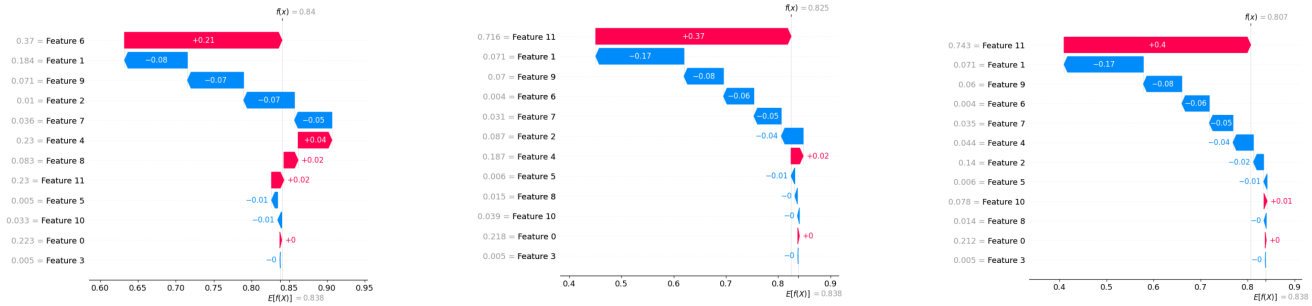
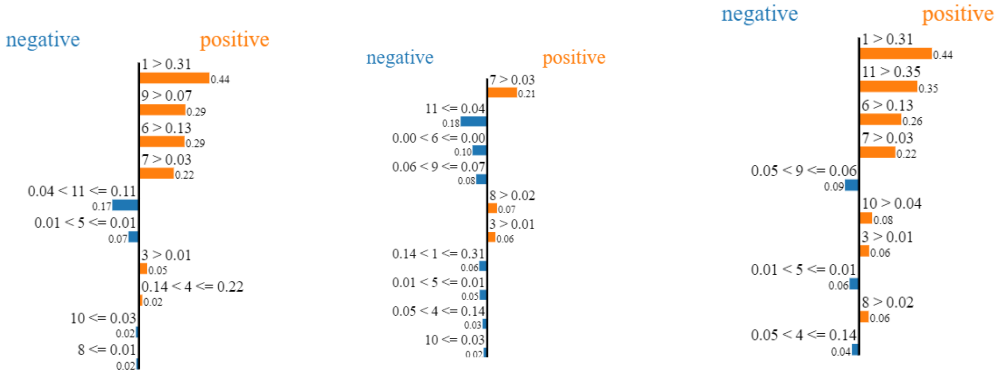


(a) (b) (c)

Fig. 3: Comparison of SHAP interpretations



(a) (b) (c)

Fig. 4: Comparison of LIME interpretations

enhancing user trust and understanding of AI-driven insights, enabling homeowners to make informed decisions about their energy consumption. By leveraging Edge Computing, we can process data locally, reducing latency and bandwidth usage while ensuring real-time analytics that is crucial for effective energy monitoring and management. ANN is a key technology used for prediction in this system. It enables accurate modelling of energy usage patterns. The XAI methods of SHAP and LIME provide an understanding of the prediction results. It is important to combine high accuracy and explainability because both users and energy providers must know the important determinants of power consumption to make reasonable decisions. More specifically, it improves real time data analysis in the architecture through the integration of edge computing for immediate feedback of energy usage patterns. This makes it free from the use of the clouds hence giving better results on the privacy and the response time. With the incorporation of ANN it is possible to load huge data in the system and get accurate predictions aimed to enhance the utilization of energy. Future research should explore the scalability of this *CHEFS* framework across various smart home devices and its adaptability to different energy sources.

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